



1 Create a NumPy Array Quickly

```
import numpy as np
arr = np.array([1, 2, 3, 4, 5])
print(arr)
```

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```
[1 2 3 4 5]
```

2 Create Arrays of Zeros & Ones

```
zeros = np.zeros((2,3))
ones = np.ones((2,3))
print(zeros)
print(ones)
```

```
[[0. 0. 0.]
 [0. 0. 0.]]
[[1. 1. 1.]
 [1. 1. 1.]]
```

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3. Generate a Range of Numbers

```
arr = np.arange(0, 10, 2)
print(arr)
```

```
[0 2 4 6 8]
```

4. Generate Random Numbers

```
rand = np.random.rand(2,3) @ clcoding
print(rand)
```

```
[[0.18587119 0.15975063 0.46365809]
 [0.11063745 0.42776541 0.32019887]]
```

5. Reshape Arrays

```
arr = np.arange(1, 10).reshape(3,3)
print(arr)
```

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
```



6. Basic Statistics (Mean, Std, Sum)

```
arr = np.array([10,20,30,40,50])  
print(arr.mean(), arr.std(), arr.sum())
```

```
30.0 14.142135623730951 150
```

7. Element-wise Operations

```
a = np.array([1,2,3])  
b = np.array([4,5,6])  
print(a + b)  
print(a * b)
```

```
[5 7 9]  
[ 4 10 18]
```

8. Boolean Indexing

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```
arr = np.array([5,10,15,20,25])  
print(arr[arr > 10])
```

```
[15 20 25]
```

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9. Matrix Multiplication

```
a = np.array([[1,2],[3,4]])  
b = np.array([[5,6],[7,8]])  
print(np.dot(a, b))
```

```
[[19 22]  
 [43 50]]
```

10. Flatten an Array

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```
arr = np.array([[1,2],[3,4]])  
print(arr.flatten())
```

```
[1 2 3 4]
```

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